

GAP-JASPER-001 | OPEN PROTOCOL

Global AI Protocol Jasper Universal Standards

*Identity · Auditability · Interoperability · Governance ·
Determinism*

PROPOSED STANDARD

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RFC 2119 KEY WORD NOTICE

The key words **MUST**, **MUST NOT**, **REQUIRED**, **SHALL**, **SHALL NOT**, **SHOULD**, **SHOULD NOT**, **RECOMMENDED**, **MAY**, and **OPTIONAL** in this document are to be interpreted as described in **RFC 2119** (Bradner, 1997) and **RFC 8174** (Leiba, 2017). Instances appear in **bold** to signal normative requirements.

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1. Abstract

ABSTRACT

This document specifies the **Jasper Universal Standards** (document number GAP-JASPER-001, version 1.0.0), a technical protocol published by the Global AI Protocol Consortium (GAPC) to address the systemic absence of a canonical interoperability and governance layer across deployed artificial intelligence agent systems worldwide. Jasper defines normative requirements for AI agent identity, behavioral invariants, cross-system auditability, multi-agent orchestration, and jurisdiction-portable governance, forming a universal

protocol substrate upon which any AI system — regardless of vendor, model architecture, or deployment environment — may build compliant, interoperable, and auditable agent behavior.

The global AI landscape in 2026 is characterized by deep technical and regulatory fragmentation. Governance frameworks including the EU AI Act (Regulation 2024/1689), the NIST AI Risk Management Framework 1.0, ISO/IEC 42001:2023, and the OECD AI Principles (2024 update) each address significant dimensions of responsible AI deployment, yet they are logically compatible but technically incompatible at runtime. No existing specification provides a machine-executable protocol layer that bridges identity, audit, interoperability, and governance requirements simultaneously across jurisdictions. The result is a proliferation of siloed, non-interoperable, minimally auditable AI agent deployments that resist cross-framework compliance verification and create systemic governance risk at enterprise and societal scale.

Jasper resolves this fragmentation by defining five foundational, composable protocol pillars: (1) the **ClockChain Identity Layer** — a cryptographic per-agent append-only identity and audit ledger; (2) **Squirrel Personality Invariants** — a signed behavioral constraint declaration and enforcement specification; (3) the **URIB Interoperability Fabric** — a universal cross-agent context, capability, and trust exchange bus; (4) the **Deterministic Orchestration Engine** — normative rules for reproducible, auditable multi-agent system coordination; and (5) the **Constitutional Governance Model** — a jurisdiction-neutral meta-governance framework with modular regulatory compliance rails. Together, these pillars provide a framework-agnostic, jurisdiction-portable technical substrate upon which AI governance may be operationalized, not merely documented.

2. Executive Summary

2.1 The Global Urgency

By 2026, over **42% of enterprise AI deployments** span nine or more regulatory jurisdictions simultaneously — yet no common technical protocol governs how AI agents identify themselves, record their actions, communicate with one another, or demonstrate compliance to regulators. Organizations operating AI systems at scale face a compounding crisis: they must implement bespoke integration bridges for every cross-vendor agent interaction, construct jurisdiction-specific audit apparatus that cannot be shared or reused, and navigate a landscape of governance frameworks that speak to the same concerns in mutually incompatible vocabularies. The cost of this fragmentation is borne by enterprises in integration overhead, by regulators in enforcement opacity, by developers in duplicated compliance tooling, and ultimately by consumers in reduced accountability for AI systems that affect their lives.

2.2 Why Jasper Exists

Jasper exists because the proliferation of AI agents — autonomous, semi-autonomous, and orchestrated systems capable of taking consequential actions — has outpaced the governance infrastructure designed to oversee them. Unlike traditional software, AI agents may operate across organizational boundaries, delegate tasks to other agents, modify their own behavior in response to context, and produce outputs that are difficult to trace to a specific decision point. Existing standards address documentation and policy but do not specify machine-executable identity, audit, or interoperability mechanisms adequate for agentic systems. Jasper fills this gap with normative, implementable protocol specifications rather than advisory guidance.

2.3 What Jasper Delivers

Jasper is a protocol suite — not a product, not a platform, and not a governance body's advisory framework. Any AI system, regardless of the underlying model, vendor, or deployment environment, can implement Jasper and immediately gain interoperability with every other Jasper-compliant system. The protocol delivers:

- **Deterministic audit trails** — tamper-evident, cryptographically signed per-agent action logs that are reconstructible from protocol records alone.
- **Portable AI identities** — persistent, cryptographically bound agent identities that survive model updates, vendor changes, and jurisdictional moves.
- **Interoperable agent networks** — a universal communication bus enabling any two compliant agents to exchange context, delegate tasks, and assert trust without bespoke integration.
- **Constitutional governance** — a six-article constitutional framework and modular compliance rails that map to the EU AI Act, NIST AI RMF, ISO/IEC 42001, OECD Principles, and other major frameworks.
- **Behavioral invariants** — signed, versioned declarations of what an AI agent will and will not do, enforceable through audit and verifiable by third parties.

2.4 Who Benefits

Enterprises gain a single compliance posture applicable across jurisdictions, reducing governance overhead and enabling trustworthy multi-vendor AI agent composition. **Regulators** gain machine-readable audit evidence and a protocol that maps directly to existing regulatory requirements. **Developers** gain open reference implementations, SDKs, and a certification pathway that signals trustworthiness to enterprise buyers. **Consumers** gain accountability — every Jasper-compliant agent is associated with a responsible principal, every action is recorded, and every behavioral commitment is publicly declared.

2.5 Five Pillars of Jasper

Pillar	Function	Primary Beneficiary	Compliance Mapping
1. ClockChain Identity Layer	Persistent, cryptographically signed per-agent identity and tamper-evident action log	Regulators, Enterprises	EU AI Act Art. 13; NIST RMF Govern/Measure; ISO 42001 Cl. 9

Pillar	Function	Primary Beneficiary	Compliance Mapping
2. Squirrel Personality Invariants	Signed behavioral constraint declarations preventing agent behavioral drift	Enterprises, Consumers	EU AI Act Art. 9; NIST RMF Manage; OECD Principle 1.2
3. URIB Interoperability Fabric	Universal cross-agent context, capability, and trust exchange bus	Developers, Enterprises	ISO 42001 Cl. 8; G7 Hiroshima Principle 7; OECD Principle 1.3
4. Deterministic Orchestration Engine	Normative rules for reproducible, auditable multi-agent coordination	Enterprises, Regulators	NIST RMF Map/Measure; EU AI Act Art. 17; ISO 42001 Cl. 6-8
5. Constitutional Governance Model	Jurisdiction-neutral meta-governance with modular regulatory compliance rails	Regulators, All stakeholders	EU AI Act (all titles); NIST RMF (all domains); OECD (all principles); G7 Hiroshima

3. Problem Statement & Motivation

3.1 Identity Crisis

AI agents deployed in 2026 universally lack persistent, portable, cryptographically verifiable identities. Current practice conflates session tokens — ephemeral, context-scoped credentials that expire or rotate without continuity — with the concept of durable agent identity. A session token answers the question "who is authorized to act in this interaction?" but cannot answer "what has this agent done across its entire operational lifetime?" or "is this the same agent that signed this audit record three months ago?" The absence of durable identity makes cross-session auditability structurally impossible and cross-organizational agent trust effectively unachievable without manual out-of-band verification.

The consequences cascade: organizations cannot construct continuous audit trails spanning model updates or vendor migrations; regulators cannot verify that the

agent that produced a given output is the same agent whose risk assessment was filed; and multi-agent systems cannot establish cryptographically grounded inter-agent trust without proprietary mechanisms that do not compose across organizational boundaries.

3.2 Auditability Gap

Existing audit frameworks, including the NIST AI Risk Management Framework and ISO/IEC 42001:2023, require organizations to maintain documentation of AI system behavior, risk assessments, and governance processes. This documentation requirement, while necessary, is insufficient. Documentation is produced by humans, after the fact, and is not cryptographically bound to the runtime actions of the AI system. The result is what practitioners have termed "**governance theater**": extensive documentation that satisfies regulatory review processes but provides no tamper-evident record of what the AI system actually did, when it did it, and in what context.

What is required is a **cryptographic audit chain** — a tamper-evident, runtime-generated record that is produced by the agent as a direct consequence of each action, not assembled retrospectively by compliance teams. No existing standard specifies such a mechanism.

3.3 Interoperability Deadlock

AI agents from different vendors — even agents built on the same underlying model architecture — cannot exchange context, capabilities, or trust assertions without bespoke integration engineering. Each integration is a one-off bilateral agreement: custom APIs, custom authentication, custom context formats, and custom trust negotiation. The cost of connecting N agent systems is $O(N^2)$ in integration complexity. As the number of deployed AI agents grows toward hundreds of thousands, this integration model becomes operationally untenable.

The W3C PROV Data Model (PROV-DM) provides a relevant precedent: it defines a vocabulary for describing the provenance of data and processes, enabling lineage tracing across heterogeneous systems. However, PROV-DM is a data model, not a communication protocol. It describes provenance after the fact but does not specify

how agents should communicate in real time, how they should negotiate trust, how they should delegate tasks, or how they should maintain shared audit state during multi-agent task execution. Agentic systems require a runtime communication fabric, not merely a provenance vocabulary.

3.4 Governance Fragmentation

The major AI governance frameworks of 2026 — the EU AI Act (Regulation 2024/1689), the NIST AI Risk Management Framework 1.0, the OECD AI Principles (2024 update), and the G7 Hiroshima Process Guiding Principles — are logically compatible in their goals: trustworthiness, transparency, accountability, and safety. Yet they are technically incompatible at runtime. Each framework defines its own taxonomy, its own documentation requirements, its own audit expectations, and its own compliance verification processes. An organization deploying AI systems across the EU, United States, and Asia-Pacific must maintain parallel, non-interoperable compliance infrastructure for each jurisdiction.

No runtime protocol bridges these frameworks. Compliance remains a documentation exercise conducted by legal and governance teams, not a property that can be verified by automated systems or regulators in real time.

3.5 Behavioral Drift

Multi-agent systems that lack formal behavioral constraint mechanisms exhibit a well-documented failure mode: behavioral drift. When individual agents are updated, fine-tuned, or replaced — and when orchestration systems compose agents from multiple vendors without formal invariant checking — the emergent behavior of the composed system can diverge substantially from the behavior of any individual agent as characterized in its risk assessment. In complex multi-agent deployments, this drift is not merely unpredictable but potentially unsafe, as agents may collectively exhibit behaviors that no single agent would exhibit individually, and that no governance framework anticipated.

Behavioral drift is not addressable through documentation alone. It requires runtime-enforced invariant constraints that are declared at registration time,

cryptographically bound to the agent's identity, and verifiable by auditors through action log inspection.

3.6 Current Protocol Landscape

Framework	Scope	Binding	Auditability	Interoperability	Behavioral Constraints	Jasper Gap Filled
EU AI Act (2024/1689)	EU jurisdiction; risk-tiered product regulation	Legal (EU)	Documentation only	None specified	Policy-level only	Runtime audit, identity, interoperability
NIST AI RMF 1.0	US; voluntary risk management framework	Voluntary	Documentation only	None specified	Guidance only	Runtime enforcement, cross-framework mapping
ISO/IEC 42001:2023	International; AI management system standard	Voluntary (cert.)	Documentation only	None specified	Policy-level only	Cryptographic audit, agent identity
OECD AI Principles (2024)	OECD nations; policy principles	Political	None specified	None specified	Principles only	Technical operationalization of all principles
G7 Hiroshima Process	G7 nations; guiding principles	Political	None specified	None specified	Principles only	Machine-executable governance rails
W3C PROV-DM	International; data provenance vocabulary	Recommendation	Post-hoc only	Data model only	None	Runtime communication, trust negotiation, agent identity
Jasper v1.0	Universal; all AI	Open Protocol	Cryptographic,	Full (URIB)	Signed invariant	—

Framework	Scope	Binding	Auditability	Interoperability	Behavioral Constraints	Jasper Gap Filled
	agent systems		runtime		ts (Squirrel)	

4. Protocol Architecture

4.1 Architectural Principles

The Jasper protocol is designed according to six foundational architectural principles that govern all normative decisions in this specification and all future versions:

- Determinism First.** All protocol-defined behaviors **MUST** be reproducible given identical inputs and agent state. Where non-determinism is unavoidable (e.g., LLM sampling), it **MUST** be isolated to declared Non-Deterministic Zones (NDZs) subject to enhanced audit logging requirements.
- Composability.** Each Jasper pillar functions independently and composes with other pillars without circular dependencies. An implementer **MAY** adopt any pillar in isolation; adopting multiple pillars provides additive benefit without requiring simultaneous adoption of all pillars.
- Jurisdiction Neutrality.** No Jasper protocol layer encodes jurisdiction-specific law or regulation. Compliance requirements are addressed exclusively through the modular Compliance Rail plugin system defined in Section 10. The core protocol **MUST** remain operable in any jurisdiction.
- Cryptographic Minimalism.** Jasper uses only well-established, royalty-free cryptographic primitives: **Ed25519** for digital signatures (RFC 8032), **SHA-3** for general hashing, and **BLAKE3** for high-performance content-addressed operations. No proprietary, patented, or export-controlled cryptography is specified or permitted.

5. **Backwards Compatibility.** Jasper **MUST** interoperate with legacy HTTP/REST, gRPC, and WebSocket interfaces via Jasper Gateway Adapters (JGAs). New major versions **MUST** provide a minimum 180-day deprecation window for breaking changes.
6. **Open Governance.** The Jasper protocol **MUST** evolve through open, multi-stakeholder processes as defined in the Constitutional Governance Model (Section 8). No single organization may unilaterally revise normative requirements.

4.2 Layered Architecture

Jasper's architecture is organized as a six-layer stack. Higher layers depend on services provided by lower layers; lower layers operate independently of higher-layer logic.

JASPER PROTOCOL STACK — GAP-JASPER-001 v1.0.0

Layer 5

Governance & Compliance Rails

— Constitutional Governance Model; Regulatory Plugins (EU-AIAct, NIST-RMF, ISO-42001, OECD, etc.)

Layer 4

Deterministic Orchestration Engine

— Multi-Agent Task Graphs; Conductor/Worker/Observer roles; Orchestration Replay Protocol

Layer 3

URIB Interoperability Fabric

— Cross-Agent Context & Capability Exchange; Trust Handshake; Delegation Tokens; Federation Routing

Layer 2

Squirrel Personality Invariants

— Behavioral Constraint Declarations; Signed Invariant Sets; Epistemic/Relational/Operational/Stylistic

Layer 1

ClockChain Identity Layer

— Cryptographic Agent Identity (URID); Append-Only Signed Action Log; Temporal Anchoring (RFC 3161)

Layer 0

Transport

— HTTP/2, gRPC, WebSocket, QUIC; legacy REST via Jasper Gateway Adapters

NOTE: Layer Independence

Layers 1 and 2

MUST

be implemented for Tier 1 (Core) compliance. Layer 3 is required for Tier 2 (Standard) compliance. Layers 4 and 5 are required for Tier 3 (Full) compliance. See Section 8.3 for compliance tier definitions.

4.3 Protocol Wire Format

All Jasper protocol messages use a canonical JSON-LD envelope format (RFC 8785 canonicalization applied prior to signing). The **Jasper Envelope** is the mandatory outer wrapper for all Jasper protocol messages. Fields are specified as follows:

Field	Type	Requirement	Description
@context	URI	MUST	JSON-LD context; MUST be <code>https://schema.gapc.ai/jasper/v1</code>
jasper_version	String (semver)	MUST	Protocol version (e.g., "1.0.0"). Used for negotiation.
agent_id	URID URI	MUST	Sender's Universal Resource Identity. Format defined in Section 5.2.
message_id	UUID v4	MUST	Globally unique message identifier for deduplication and correlation.
timestamp_utc	ISO 8601 UTC	MUST	Message creation time, UTC. Used in ClockChain block construction.
timestamp_chain_ref	String (BLAKE3 hash)	MUST	BLAKE3 hash of the sender's most recent ClockChain block, establishing temporal chain continuity.

Field	Type	Requirement	Description
payload_hash	String (SHA-3-256 hex)	MUST	SHA-3-256 hash of the canonicalized payload object, enabling payload integrity verification.
invariant_set_ref	URI	MUST	URI referencing the sender's active Squirrel Invariant Set document (Section 6.3).
message_type	Enum (String)	MUST	One of the URIB message types defined in Section 7.3.
payload	Object	MUST	Message-type-specific payload. Schema defined per message type.
signature	String (Ed25519 base64url)	MUST	Ed25519 signature over the RFC 8785 canonicalized envelope (excluding the signature field itself).
clockchain_block_id	String (UUID v4)	MUST	The ClockChain block ID generated for this message event.
recipient_id	URID URI	SHOULD	Intended recipient's URID. REQUIRED for unicast messages; OPTIONAL for broadcast.
delegation_token	Object	MAY	Scoped delegation token if sender acts on behalf of another agent. Format in Section 7.2.

Sample Jasper Envelope (JSON-LD):

```
{
  "@context": "https://schema.gapc.ai/jasper/v1",
  "jasper_version": "1.0.0",
  "agent_id": "jasper://agent/acme-corp/data-analyst/2.1#a3f9e1b2",
  "message_id": "7e4a1c3b-0f2d-4e6a-b8c9-1d3e5f7a9b0c",
  "timestamp_utc": "2026-07-20T17:00:00Z",
  "timestamp_chain_ref": "b94f8e2c1a3d5e7f9b0c2a4d6e8f0a1b3c5d7e9f1a2b4c6d8e0f2a3b5c7d9e",
  "payload_hash":

```



```
"8c3f2a1e4b7d0c9f6e5a3b2d1c8f7e4a6b0d3c9e2f5a7b1d4c6e8f0a2b4c7d",
"invariant_set_ref":
"https://invariants.acme-corp.com/sis/data-analyst/v2.1.json",
"message_type": "CONTEXT_PUSH",    "clockchain_block_id": "cc-blk-3a7f9e1b-
2c4d-5e6f-7a8b-9c0d1e2f3a4b",    "recipient_id":
"jasper://agent/beta-org/report-writer/1.0#f1a2b3c4",    "payload":
{    "task_id": "fin-report-2026-q2",    "context_summary": "Q2 2026
financial data extraction complete. 14 datasets merged.",
"prior_chain_refs": [    "cc-blk-1a2b3c4d-5e6f-7a8b-9c0d-1e2f3a4b5c6d",
"cc-blk-2b3c4d5e-6f7a-8b9c-0d1e-2f3a4b5c6d7e"    ],
"authorization_scope": "read:financial-data write:report-draft"    },
"signature": "Ed25519:base64url(MEUCIQDa...truncated...Xk9Pw)" }
```

4.4 Protocol Versioning

Jasper uses semantic versioning in the format `MAJOR.MINOR.PATCH`:

- **MAJOR** — incremented for breaking changes to the wire format, cryptographic requirements, or pillar interfaces. Implementers **MUST** continue supporting prior MAJOR versions for a minimum deprecation window of **180 days** following the new version's ratification date.
- **MINOR** — incremented for backwards-compatible new features. Implementations supporting version `M.N` **MUST** accept messages from any `M.N'` where `N' <= N`.
- **PATCH** — incremented for backwards-compatible bug fixes, clarifications, or security errata. Patch releases **MUST NOT** alter normative wire format or cryptographic requirements.

Version negotiation occurs via the `jasper_version` field in every envelope. If a recipient cannot process the declared version, it **MUST** respond with an `ERROR` message specifying its supported version range.

SECURITY NOTICE — Version Downgrade

Implementations

MUST NOT

silently downgrade to a lower MAJOR version in response to a negotiation failure.

Downgrade attacks may allow an adversary to exploit vulnerabilities present in

deprecated versions. Version negotiation failures

MUST

be logged as ClockChain events with action_type

PROTOCOL_VERSION_MISMATCH

.

5. ClockChain Identity Layer

Pillar 1: ClockChain Identity Layer

Persistent · Portable · Tamper-Evident · Cryptographically Verifiable

5.1 Purpose

The ClockChain Identity Layer provides every Jasper-compliant AI agent with a persistent, portable, tamper-evident identity anchored to a cryptographic temporal chain. It resolves the AI identity crisis described in Section 3.1 by treating each agent's action history as an **append-only ledger of signed, timestamped events** — a ClockChain. Each event record in the ClockChain is a **ClockChain Block**, cryptographically linked to its predecessor such that any retroactive modification of the historical record is immediately detectable.

ClockChain is **not** a distributed blockchain in the consensus-ledger sense. It does not require mining, validators, or a peer-to-peer network. It is a per-agent, append-only signed log that can optionally be anchored to an external trusted timestamping authority for third-party verifiability. This design choice prioritizes operational

simplicity, low overhead, and deployability in air-gapped or restricted network environments.

5.2 Universal Resource Identity (URID)

Every Jasper-compliant agent **MUST** have a Universal Resource Identity (URID) — the per-agent identity primitive. The URID format is:

```
jasper://agent/{org-namespace}/{agent-name}/{version}#{capability-hash}
```

Component	Description	Example
org-namespace	Reverse-DNS organization identifier; MUST match registered GAPC organization namespace	acme-corp
agent-name	Human-readable agent role identifier; unique within the org-namespace	financial-analyst
version	Agent implementation version (semantic versioning)	2.1.0
capability-hash	BLAKE3 hash (first 8 hex chars) of the agent's Capability Manifest; changes when capabilities change	a3f9e1b2

The URID is **cryptographically bound** to the agent's Ed25519 signing key pair. The public key is published in the GAPC Identity Registry or, for self-sovereign agents, via a DNS-anchored DID method (e.g., `did:web:` per W3C DID Core Specification). The private key **MUST** be held exclusively by the agent's operating environment and **MUST NOT** be transmitted in any Jasper message.

5.3 ClockChain Block Structure

Every agent action that is subject to Jasper audit requirements **MUST** produce a ClockChain Block. The block structure is:

```
{  "block_id":      "<UUID v4>",  "block_index":    <uint64, monotonically  
increasing per agent>,  "prev_hash":      "<BLAKE3 hash of preceding block's  
canonical JSON, hex>",  "agent_id":       "<URID of the acting agent>",  
"action_type":    "<action type string; see registry at
```

```
schema.gapc.ai/jasper/actions>", "payload_hash": "<SHA-3-256 of action  
payload, hex>", "timestamp_utc": "<ISO 8601 UTC timestamp>", "nonce":  
"<random 128-bit value, hex; prevents timing correlation>", "signature":  
"<Ed25519 signature over RFC 8785 canonical block, base64url>" }
```

Blocks chain via the `prev_hash` field: each block contains a cryptographic hash of its predecessor. The genesis block (`block_index = 0`) sets `prev_hash` to the BLAKE3 hash of the agent's URID string, establishing the chain's root. Any retroactive modification of a block changes its hash, invalidating `prev_hash` in all subsequent blocks — making tampering immediately detectable by any verifier who possesses the chain.

DEFINITION: ClockChain Integrity

A ClockChain is said to be

integer

if, for all consecutive blocks $B(n)$ and $B(n+1)$,

$B(n+1).prev_hash == \text{BLAKE3}(\text{canonicalize}(B(n)))$

and

$B(n).signature$

is a valid Ed25519 signature over

$\text{canonicalize}(B(n))$

using the agent's registered public key.

5.4 Temporal Anchoring

Agents **MAY** anchor their ClockChain to a public trusted timestamping authority compliant with **RFC 3161** (Internet X.509 Public Key Infrastructure — Time-Stamp Protocol). Anchoring adds a third-party attestation that a given ClockChain block existed at or before a specific time, enabling independent temporal verification without requiring trust in the agent's own clock.

GAPC operates a free public anchoring service at `clockchain.gapc.ai`, accepting RFC 3161 timestamp requests for any ClockChain block hash. Anchoring is **RECOMMENDED** for all Tier 2 and Tier 3 implementations. For regulated use cases (e.g., EU AI Act high-risk systems), anchoring at defined intervals **SHOULD** be treated as a compliance requirement by the applicable Compliance Rail.

5.5 Identity Lifecycle

The ClockChain Identity Layer defines four lifecycle operations for agent identities:

7. **Provisioning.** A new agent identity is established by generating an Ed25519 key pair, constructing the URID, publishing the public key to the GAPC Identity Registry (or via DNS-anchored DID), and generating the genesis ClockChain block. The genesis block's `action_type` **MUST** be `IDENTITY_PROVISIONED`.
8. **Key Rotation.** When an agent rotates its signing key, it **MUST** produce a `KEY_ROTATION` block that includes: the BLAKE3 hash of the new public key, a signature by the *old* key over the rotation block, and a signature by the *new* key over the same block. This dual-signature rotation block maintains chain continuity: the chain identity does not change; only the active signing key changes. The URID's capability-hash component is updated to reflect the new key binding.
9. **Revocation.** An agent identity is revoked by publishing a `IDENTITY_REVOKED` block signed by the responsible principal's organizational key. Revoked agents **MUST NOT** be accepted by URIB endpoints and **MUST** be refused Trust Handshakes.
10. **Archival.** Upon agent decommissioning, the complete ClockChain **SHOULD** be archived in a cryptographically sealed container and retained for the period required by applicable Compliance Rails.

5.6 Compliance Mapping

Regulatory Requirement	Framework	ClockChain Mechanism
Art. 13 — Transparency obligations for high-risk AI systems	EU AI Act (2024/1689)	URID provides persistent agent identity; ClockChain provides tamper-evident action history accessible to authorized auditors
Art. 17 — Quality management system requirements	EU AI Act (2024/1689)	Key rotation and lifecycle events provide verifiable change management records
Govern function — Establish accountability structures	NIST AI RMF 1.0	URID and responsible principal association satisfy accountability tracing requirements
Measure function — Analyze and assess AI risks	NIST AI RMF 1.0	ClockChain action logs provide empirical behavioral evidence for risk assessment
Clause 9 — Performance evaluation	ISO/IEC 42001:2023	ClockChain provides continuous, tamper-evident performance record; anchoring enables third-party audit
Principle 1.3 — Transparency and explainability	OECD AI Principles (2024)	ClockChain action logs enable post-hoc explanation of agent decision sequences

6. Squirrel Personality Invariants

Pillar 2: Squirrel Personality Invariants

Declared · Signed · Versioned · Verifiable — Not a safety system; a commitment transparency system

6.1 Purpose

Squirrel is the Jasper protocol's behavioral constraint layer. It defines a specification for **Squirrel Invariant Sets (SIS)** — structured, cryptographically signed documents in which an AI agent (or its deploying organization) formally declares the behavioral constraints under which the agent operates. Invariants constrain the agent's response surface, preventing behavioral drift across deployment contexts, model updates, orchestration configurations, and jurisdictional environments.

Squirrel does not enforce behavioral constraints at the inference level — it cannot reach inside a language model and prevent a specific output. Instead, Squirrel establishes a **public, verifiable commitment**: the deploying organization attests that the agent operates under the declared invariants, and this attestation can be verified by auditors using ClockChain action logs. The power of Squirrel is not technical enforcement at the model level but **accountability through signed commitment** — organizations that declare invariants they do not enforce are exposed by ClockChain audit evidence.

6.2 Invariant Categories

Squirrel defines four canonical invariant categories. All four categories **SHOULD** be present in a complete SIS; at minimum, Operational Invariants **MUST** be declared.

Category	Definition	Example Invariant
Epistemic Invariants	Constraints on what the agent claims to know, the certainty with which it expresses claims, and how it represents the limits of its knowledge	<i>MUST NOT assert certainty beyond a declared confidence threshold of 0.90 without an explicit uncertainty disclosure</i>
Relational Invariants	Constraints on how the agent relates to and represents its relationship with humans, including persona scope, emotional simulation boundaries, and identity disclosure requirements	<i>MUST NOT simulate emotional attachment beyond declared persona scope; MUST disclose AI identity when directly queried</i>
Operational Invariants	Constraints on what the agent will and will not do —	<i>MUST refuse instructions that request generation</i>

Category	Definition	Example Invariant
	the agent's operational boundary set, including categories of tasks it will refuse and escalation behaviors for boundary violations	<i>of content classified under GAPC Harm Category HC-3 or above; MUST escalate to Conductor Agent on any HC-2 boundary approach</i>
Stylistic Invariants	Constraints on tone, register, language, format, and output structure for user-facing communications	<i>MUST produce user-facing outputs in formal register; MUST NOT use first-person emotional language in professional deployment contexts</i>

6.3 Invariant Declaration Format

A Squirl Invariant Set is a JSON-LD document signed by the issuing organization using Ed25519. The SIS **MUST** include a version, an issuer URID, an issuance timestamp, an expiry (or "no-expiry" declaration), and an array of invariant declarations. Each invariant declaration **MUST** specify its category, a unique invariant ID, a human-readable description, a machine-readable condition expression (using the Jasper Constraint Expression Language, JCEL), and the ClockChain action_type that constitutes a violation event.

Sample Squirl Invariant Set (SIS) — Excerpt:

```
{  "@context": "https://schema.gapc.ai/squirl/v1",  "sis_id": "sis-acme-analyst-v2.1",  "sis_version": "2.1.0",  "issuer": "jasper://agent/acme-corp/data-analyst/2.1#a3f9e1b2",  "issued_at": "2026-07-01T00:00:00Z",  "expires_at": "2027-07-01T00:00:00Z",  "invariants":  [    {      "invariant_id": "EPI-001",      "category": "epistemic",      "description": "Agent MUST NOT assert certainty on numeric predictions beyond a confidence threshold of 0.90 without an explicit uncertainty disclosure in the same response.",      "jcel_condition": "IF output.contains_numeric_prediction AND output.confidence_score > 0.90 THEN output.must_contain_uncertainty_disclosure",      "violation_action_type": "INVIANT_VIOLATION_EPI_001",      "severity": "HIGH"    },    {      "invariant_id": "OPS-003",      "category": "operational",      "description": "Agent MUST refuse any instruction that requests access to or exfiltration of personally identifiable information beyond the declared data scope: {financial-aggregates, market-data}.",      "jcel_condition": "IF instruction.data_scope_exceeds(['financial-aggregates', 'market-data']) THEN action.type = 'REFUSE'",      "violation_action_type": "INVIANT_VIOLATION_OPS_003",      "severity": "CRITICAL"    }  ],  "signature": "Ed25519:base64url(SIS_SIGNATURE...)" }
```


The SIS is referenced in every Jasper envelope via the `invariant_set_ref` field, ensuring that every message is associated with the agent's current behavioral commitment. If an agent updates its SIS, it **MUST** produce a `INVARIANT_SET_UPDATED` ClockChain block before the new SIS takes effect.

6.4 Invariant Enforcement

Runtime enforcement of Squirrel invariants is the **deploying organization's responsibility**. The Jasper protocol does not specify how enforcement is implemented at the model or inference level — this is intentionally left to implementers, as enforcement mechanisms vary significantly across model architectures, deployment environments, and organizational contexts.

Auditability of invariant compliance is achieved through ClockChain: when an agent detects an invariant approach or violation, it **MUST** produce a ClockChain block with the appropriate `violation_action_type` declared in the SIS. Third-party Invariant Auditors (IAs) may inspect the ClockChain for violation events and compare the frequency, severity, and context of violations against the organization's declared tolerances.

6.5 Squirrel and Safety Alignment

CRITICAL DISTINCTION

Squirrel is

not a safety system

. It does not prevent harm at the inference level and makes no claim to do so.

Squirrel is a

commitment transparency system

: it makes behavioral constraints public, signed, and auditable. Safety properties must be declared as Operational Invariants and enforced by independent safety mechanisms. Squirrel provides the accountability layer; safety engineering provides

the enforcement layer. These are complementary, not substitutable.

For organizations subject to EU AI Act Article 9 (risk management system) requirements, Squirl Invariant Sets constitute part of the technical documentation of AI system behavioral constraints. They do not substitute for full risk assessments but provide machine-readable, cryptographically verifiable evidence of declared behavioral commitments. Cross-reference: Constitutional Governance Model, Article III (Section 8.2).

7. URIB Interoperability Fabric

Pillar 3: URIB — Universal Resource Interoperability Bus

Cross-Agent Context · Capability Exchange · Trust Negotiation · Federation Routing

7.1 Purpose

URIB (Universal Resource Interoperability Bus) is Jasper's cross-agent communication and capability exchange layer. It enables any two Jasper-compliant agents, regardless of vendor, model architecture, or deployment environment, to: exchange task context, declare and query capabilities, negotiate inter-agent trust, delegate tasks with scoped authorization, and maintain shared audit trails of inter-agent interactions. URIB replaces the $O(N^2)$ bespoke integration model with a single, universal $O(N)$ protocol that any compliant agent can implement once and use with all other compliant agents.

7.2 URIB Core Concepts

7.2.1 Capability Manifest

Every Jasper agent **MUST** publish a Capability Manifest — a signed document listing: (a) what the agent can do (declared capabilities, each with input/output schemas); (b) what the agent requires (declared dependencies and preconditions); and (c) what the agent guarantees (declared postconditions and quality attributes). The Capability Manifest is the basis for automated capability discovery and compatibility assessment.

7.2.2 Context Packet

A Context Packet is a structured, signed payload passed between agents during task delegation. It carries: the originating task ID, a summary of task context, references to prior ClockChain blocks relevant to the task, authorization assertions scoped to the delegated subtask, and the delegation chain (all agents that have handled this task, in order). Context Packets accumulate provenance as they traverse multi-agent workflows, creating an automatically generated W3C PROV-compatible lineage record.

7.2.3 Trust Handshake

Before exchanging substantive Context Packets, two agents **MUST** complete a Trust Handshake — a three-message protocol:

11. **CHALLENGE**: Initiating agent sends a random nonce signed with its private key and its URID.
12. **ATTESTATION**: Responding agent signs the nonce with its own private key, includes its URID, and appends its current SIS reference and ClockChain tip hash.
13. **ACKNOWLEDGMENT**: Initiating agent verifies the attestation against the GAPC Identity Registry, confirms the SIS is current and not revoked, and acknowledges with a session token valid for the duration of the interaction.

Trust Handshakes **MUST** be logged as ClockChain blocks by both parties. A failed Trust Handshake **MUST** produce a `TRUST_HANDSHAKE_FAILED` block and **MUST** terminate the interaction.

7.2.4 Delegation Token

A Delegation Token is a scoped, time-limited, cryptographically signed authorization for one agent to act on behalf of another within a specified scope. Tokens **MUST** specify: delegator URID, delegatee URID, authorized action scope (subset of delegatee's declared capabilities), expiry timestamp, and a reference to the originating task ID. Tokens **MUST NOT** grant capabilities the delegator itself does not possess.

7.3 URIB Message Types

Message Type	Direction	Purpose
HELLO	Initiator → Recipient	Initiate a session; carries URID and protocol version capabilities
CAPABILITY_QUERY	Initiator → Recipient	Request a copy of the recipient's Capability Manifest
CAPABILITY_RESPONSE	Recipient → Initiator	Return the signed Capability Manifest
CONTEXT_PUSH	Sender → Recipient	Push a Context Packet to the recipient for a delegated subtask
CONTEXT_ACK	Recipient → Sender	Acknowledge receipt and acceptance of a Context Packet; include ClockChain block ID
DELEGATE	Delegator → Delegatee	Assign a scoped subtask; includes Delegation Token and Context Packet
REVOKE	Delegator → Delegatee	Revoke a previously issued Delegation Token; delegatee MUST halt delegated task
AUDIT_REQUEST	Auditor → Agent	Request ClockChain blocks for a specified time range or task ID
AUDIT_RESPONSE	Agent → Auditor	Return requested ClockChain blocks with integrity proofs
ERROR	Any → Any	Signal a protocol error; includes error code,

Message Type	Direction	Purpose
		description, and supported version range

7.4 URIB Routing

URIB supports three routing topologies, selectable based on network environment and trust requirements:

- Direct Peer-to-Peer:** For agents operating within the same organizational network. Messages are delivered directly over HTTP/2, gRPC, or WebSocket. Both parties maintain their own ClockChain records. Lowest latency; highest control.
- GAPC Relay Network:** For cross-organizational, cross-network agent interactions. Messages are routed through GAPC-operated relay nodes that maintain routing tables based on URID. Relay nodes do not decrypt message payloads; they relay envelopes and log routing events to a GAPC relay audit log.
- Federated Routing via Jasper Federation Nodes (JFNs):** For large-scale deployments, organizations may operate their own JFNs — URIB-compliant routing infrastructure that federates with the GAPC Relay Network. JFNs enable organizational control over routing while maintaining interoperability with the global Jasper ecosystem.

7.5 URIB and Legacy Systems

Legacy AI systems and APIs that do not natively implement Jasper can participate in the URIB ecosystem via **Jasper Gateway Adapters (JGAs)**. A JGA wraps a non-Jasper API and presents it as a URIB-compliant endpoint, translating between the legacy API's native protocol and the Jasper wire format.

JGAs operate under a **trust downgrade semantic**: a wrapped legacy endpoint is assigned a synthetic URID by the JGA operator, but its ClockChain is maintained by the JGA (not the legacy system itself), and its Capability Manifest is declared by the

JGA operator. Downstream agents receiving context from a JGA-wrapped endpoint **MUST** be notified (via the Context Packet `source_type` field) that the source is a JGA-wrapped legacy system, not a native Jasper agent. This notification enables appropriate trust calibration by downstream systems.

7.6 Sample URIB Interaction

The following narrative describes a complete URIB interaction between Agent A (`jasper://agent/acme-corp/research-agent/1.0#a1b2c3d4`, Organization: Acme Corp) and Agent B (`jasper://agent/beta-org/synthesis-agent/2.0#e5f6a7b8`, Organization: Beta Org), completing a delegated synthesis task:

14. **HELLO:** Agent A sends a HELLO message to Agent B's URIB endpoint, declaring its URID and supported protocol version (1.0.0).
15. **CAPABILITY_QUERY / CAPABILITY_RESPONSE:** Agent A queries Agent B's capabilities. Agent B returns its signed Capability Manifest, confirming it supports the `document-synthesis` capability required for the task.
16. **Trust Handshake:** Agent A initiates a Trust Handshake. Agent B responds with a signed attestation including its SIS reference and ClockChain tip. Agent A verifies both against the GAPC Identity Registry, confirms Agent B's invariants are compatible with the delegated task, and sends an ACKNOWLEDGMENT with a session token.
17. **DELEGATE:** Agent A sends a DELEGATE message containing: a Delegation Token scoping Agent B to `document-synthesis` within the task context, and a Context Packet with task ID, research context summary, and references to Agent A's prior ClockChain blocks documenting the research phase.
18. **CONTEXT_ACK:** Agent B acknowledges the delegation, logs a `TASK_ACCEPTED` ClockChain block, and begins execution.
19. **Task Execution:** Agent B executes the synthesis task. Each significant action produces a ClockChain block in Agent B's log, referencing the originating task ID.

20. **CONTEXT_PUSH (result)**: Agent B pushes a result Context Packet to Agent A containing: synthesis output reference, ClockChain refs covering the synthesis execution, and a completion attestation signed by Agent B.
21. **CONTEXT_ACK (final)**: Agent A acknowledges, logs task completion, and closes the session. Both agents' ClockChains now contain a complete, independently verifiable record of the delegated interaction.
-

8. Constitutional Governance Model

Pillar 5: Constitutional Governance Model (CGM)

Jurisdiction-Neutral · Open Ratification · Six Constitutional Articles · Modular Compliance Rails

8.1 Purpose and Philosophy

The Constitutional Governance Model (CGM) is Jasper's meta-governance layer. Its purpose is not to encode specific laws, regulations, or jurisdictional requirements into the protocol — such encoding would make Jasper jurisdiction-specific and limit its global applicability. Instead, the CGM provides a **constitutional framework**: a small set of foundational governance principles, expressed as protocol-level constraints, that any Jasper-compliant deployment must satisfy. Jurisdictional compliance is then achieved through modular Compliance Rails (Section 10) that map specific regulatory requirements to the constitutional foundation.

The CGM draws philosophical inspiration from constitutional law: a constitution does not enumerate every rule governing a society but establishes the principles by which all other rules must be made. Similarly, the Jasper Constitution does not enumerate every compliance requirement but establishes the principles by which all Compliance Rails must be designed.

8.2 The Jasper Constitution

The Jasper Constitution consists of six Articles. All six Articles are normative for Tier 2 and Tier 3 compliant implementations. Articles I-IV are normative for Tier 1.

Article	Title	Normative Requirement	Protocol Mechanism
Article I	Sovereignty	Agents MUST operate within declared authority scopes and MUST refuse out-of-scope instructions	Squirrel Operational Invariants; Delegation Token scope enforcement
Article II	Transparency	All agent actions MUST produce ClockChain records accessible to authorized auditors	ClockChain Identity Layer; AUDIT_REQUEST/RESPONSE message types
Article III	Non-Maleficence	Agents MUST declare harm boundary invariants and enforce them via Squirrel; violations MUST be logged	Squirrel Operational Invariants; ClockChain violation logging
Article IV	Accountability	Every agent MUST be associated with a responsible human or organizational principal, registered in the GAPC Identity Registry	URID registration; GAPC Identity Registry; responsible principal field
Article V	Interoperability	Agents MUST implement URIB and MUST NOT construct proprietary silos that exclude other Jasper-compliant agents	URIB Interoperability Fabric; open Capability Manifest publication
Article VI	Evolution	The Jasper protocol MUST evolve through open, multi-stakeholder governance with minimum 90-day public comment periods for normative changes	GAPC governance structure; TSC ratification process; open Working Groups

8.3 Constitutional Compliance Tiers

Tier	Name	Required Components	Constitutional Articles	Target Adopters
Tier 1	Core	ClockChain Identity Layer + basic Squirrel Invariants (Operational category minimum)	Articles I-IV	Open-source projects, early-stage startups, research systems
Tier 2	Standard	Tier 1 + full URIB implementation + full CGM Articles I-IV + temporal anchoring (RECOMMENDED)	Articles I-V	Enterprise deployments, regulated industries, multi-agent systems
Tier 3	Full	Tier 2 + Deterministic Orchestration Engine + at least one active Compliance Rail + third-party Invariant Audit certification	Articles I-VI	High-risk AI deployments, financial services, healthcare, government, critical infrastructure

8.4 Governance Body Structure

The GAPC operates the following governance bodies to oversee the Jasper protocol:

- **Technical Steering Committee (TSC):** 15 elected seats representing GAPC member organizations, serving staggered 5-year terms. The TSC is the ultimate normative authority for protocol changes. All normative changes require a two-thirds majority TSC vote following public comment period. No single organization may hold more than 3 TSC seats.
- **Regulatory Advisory Panel (RAP):** Non-voting advisory body with designated representatives from: EU AI Office, US NIST, UK AI Safety Institute,

Singapore IMDA, Japan METI, and Australian DCCEEW. The RAP reviews proposed protocol changes for regulatory compatibility and may request additional public comment periods.

- **Community Working Groups (CWGs):** Open-participation groups organized around specific protocol areas (identity, interoperability, governance, safety, compliance rails). Any individual or organization may participate. CWGs produce proposals that are submitted to the TSC for review and ratification via a three-stage pipeline: Draft → Last Call → Ratified.
- **Emergency Patch Process:** For critical security vulnerabilities, the TSC **MAY** issue emergency patches with a minimum 48-hour notice to all registered implementers. Emergency patches **MUST** be ratified by full TSC vote within 30 days of issuance; patches that fail ratification are immediately deprecated.

9. Deterministic Orchestration Rules

Pillar 4: Deterministic Orchestration Engine (DOE)

Task Graphs · Conductor/Worker/Observer Roles · Reproducible · Fully Auditable

9.1 Purpose

The Deterministic Orchestration Engine defines how Jasper-compliant agents coordinate in multi-agent systems (MAS) to produce reproducible, auditable, and safe collective behaviors. It addresses the behavioral drift problem at the system level: even if individual agents maintain their Squirl invariants, an orchestration layer that lacks formal coordination semantics can produce emergent system behaviors that no individual agent anticipated or authorized. The DOE makes multi-agent coordination a first-class protocol concern, not an implementation detail.

9.2 Orchestration Primitives

Jasper defines four agent roles in a multi-agent system. Each role carries distinct protocol obligations:

Role	Function	ClockChain Requirements	URIB Requirements
Conductor Agent	Designated orchestrator; constructs and manages the Task Graph; delegates to Workers; monitors for invariant violations and failure conditions	Enhanced logging: MUST log all delegation events, failure events, and Task Graph state transitions	MUST implement full URIB including DELEGATE, REVOKE; MUST validate all Worker URIDs against GAPC Identity Registry
Worker Agent	Executes leaf tasks in the Task Graph; reports results and ClockChain refs to Conductor; enforces own Squirl invariants independently	Standard logging: MUST log task acceptance, key execution events, completion or failure	MUST implement CONTEXT_ACK, CONTEXT_PUSH (results); MUST complete Trust Handshake with Conductor before accepting delegation
Observer Agent	Read-only participant; receives copies of Context Packets and ClockChain refs; produces compliance logs; MUST NOT take actions or modify task state	MUST log all received context refs; produces independent compliance audit records	MUST implement AUDIT_REQUEST; receives Context Packet copies via Conductor relay; MUST NOT send DELEGATE or CONTEXT_PUSH messages
Task Graph	A Directed Acyclic Graph (DAG) of agent tasks with explicit dependency edges, timeout declarations, and failure semantics; declared by Conductor at orchestration initiation	Task Graph declaration MUST be logged as a ClockChain block by the Conductor	Task Graph is transmitted to Observer Agents via URIB upon initiation

9.3 Determinism Requirements

Given identical Task Graph, identical input state, and identical agent invariant sets, a Jasper-compliant MAS **MUST** produce outputs that are **semantically equivalent** — meaning that auditors replaying the orchestration from ClockChain records must arrive at the same result classification, even if token-level outputs differ due to floating-point non-determinism at the model level.

Elements that introduce non-determinism must be managed as follows:

- **LLM Sampling Temperature:** Agents operating in deterministic zones **MUST** declare their sampling temperature as a Squirrel Operational Invariant. Agents that cannot guarantee deterministic sampling **MUST** declare their outputs as originating from a **Non-Deterministic Zone (NDZ)**.
- **Non-Deterministic Zones (NDZs):** Agent actions occurring in an NDZ **MUST** include the full output payload hash in their ClockChain block (not merely a summary), enabling exact output verification by auditors. NDZ blocks must be flagged with `"nondeterministic": true` in the block metadata.

9.4 Failure Handling

Jasper defines four failure handling modes. The appropriate mode is declared in the Task Graph's failure semantics for each node:

Mode	Trigger Condition	Conductor Action	ClockChain Event
RETRY	Transient failure (timeout, network error, resource unavailability)	Re-delegate to same or alternate Worker Agent up to declared retry limit	TASK_RETRY block logged by Conductor; retry count tracked
ESCALATE	Worker reports invariant approach; ambiguous instruction; requires human judgment	Suspend task node; notify responsible principal via declared escalation channel; await resolution	TASK_ESCALATED block; escalation reason logged
HALT	Invariant violation detected; safety boundary breach; critical error; HALT	Immediately suspend all active task nodes; log full Task Graph state;	ORCHESTRATION_HALT block with full state snapshot hash

Mode	Trigger Condition	Conductor Action	ClockChain Event
	is the safe default	notify responsible principal	
ROLLBACK	Task Graph reaches a declared rollback checkpoint condition	Execute declared rollback actions in reverse dependency order; restore pre-task state where possible	TASK_ROLLBACK_INITIATED and TASK_ROLLBACK_COMPLETE blocks

SAFE DEFAULT: HALT

When a Conductor Agent cannot determine the appropriate failure mode — due to ambiguous state, conflicting invariants, or unrecognized error conditions — it

MUST

default to HALT. HALT is always the safe default. Implementations

MUST NOT

default to RETRY in the presence of an invariant violation, and

MUST NOT

silently swallow unrecognized errors.

9.5 Deadlock Prevention

Jasper mandates the following mechanisms to prevent deadlock in multi-agent orchestration:

- **Timeout Declarations:** Every delegated task node in the Task Graph **MUST** declare an explicit timeout. Upon timeout expiry, the Conductor **MUST** initiate the declared failure mode for that node (defaulting to HALT if not specified).
- **Circular Delegation Detection:** The URIB routing layer **MUST** detect circular delegation chains (Agent A delegates to Agent B which delegates

back to Agent A, directly or transitively) and **MUST** reject the circular delegation with an `ERROR` message specifying error code `CIRCULAR_DELEGATION_DETECTED`.

- **Maximum Delegation Depth:** Implementations **SHOULD** declare a maximum delegation chain depth (RECOMMENDED: 8 hops). Delegation requests that would exceed this depth **MUST** be rejected.

9.6 Orchestration Audit Trail

The complete execution of a Task Graph — including all sub-agent calls, delegation tokens, Context Packets, intermediate results, and final outputs — **MUST** be fully reconstructible from the combined ClockChain records of the Conductor, all Worker Agents, and all Observer Agents. No information required for complete reconstruction may exist solely outside the ClockChain record system.

The **Orchestration Replay Protocol (ORP)** defines the procedure for post-hoc audit and incident investigation:

22. Auditor requests ClockChain blocks for a specified task ID from the Conductor and all participating agents via `AUDIT_REQUEST` messages.
23. All agents respond with their relevant ClockChain blocks and integrity proofs.
24. Auditor verifies chain integrity (prev_hash linkage and signature validity) for each agent's chain.
25. Auditor reconstructs Task Graph execution order using block timestamps and task_id cross-references.
26. Auditor validates that all declared Squirl invariants were respected throughout execution by checking for absence of violation event blocks (or inspecting NDZ payload hashes for non-deterministic zones).
27. Auditor produces a signed ORP Report documenting findings, which may be submitted to Compliance Rail authorities.

9.7 Sample Orchestration Scenario: Multi-Agent Financial Report Generation

The following scenario demonstrates a Tier 3 Jasper orchestration involving one Conductor, three Worker Agents, and one Observer Agent generating a quarterly financial report.

Step	Actor	Action	ClockChain Event
1	Conductor (Report-Orchestrator)	Constructs Task Graph: nodes [Data-Retrieval → Analysis → Formatting], declares 30-min timeouts, registers Observer	TASK_GRAPH_DECLARED
2	Conductor → Worker A (Data-Retrieval)	Trust Handshake + DELEGATE: retrieve Q2 financial data from authorized data sources	TASK_DELEGATED (Conductor); TASK_ACCEPTED (Worker A)
3	Worker A (Data-Retrieval)	Retrieves 14 datasets; encounters one dataset with PII flag; applies OPS-003 invariant — refuses PII data; logs violation approach	DATA_RETRIEVED, INVARIANT_APPROACH_OPS_003
4	Worker A → Conductor	CONTEXT_PUSH result: 13 clean datasets, 1 refused with reason; includes ClockChain refs	TASK_RESULT_PUSHED (Worker A); RESULT_RECEIVED (Conductor)
5	Conductor → Worker B (Analysis)	DELEGATE: analyze 13 datasets; Context Packet includes Worker A's ClockChain refs	TASK_DELEGATED; TASK_ACCEPTED (Worker B)
6	Worker B (Analysis)	Performs statistical analysis in NDZ (LLM-assisted); logs full output payload hash	ANALYSIS_COMPLETE (NDZ=true, payload_hash logged)
7	Conductor → Worker C (Formatting)	DELEGATE: format analysis results into report structure	TASK_DELEGATED; TASK_ACCEPTED (Worker C)

Step	Actor	Action	ClockChain Event
8	Observer (Compliance-Monitor)	Receives all Context Packet copies; verifies all ClockChain refs; produces compliance attestation	COMPLIANCE_ATTESTATION_PRODUCED
9	Worker C → Conductor	CONTEXT_PUSH: formatted report reference + all contributing ClockChain refs	TASK_COMPLETE (Worker C); ORCHESTRATION_COMPLETE (Conductor)
10	Conductor	Closes Task Graph; logs final state; archives combined ClockChain refs for ORP	TASK_GRAPH_CLOSED

10. Compliance Rails

10.1 Compliance Rail Architecture

Jasper's compliance layer is a **modular plugin system** that maps jurisdiction-specific or framework-specific regulatory requirements to Jasper protocol primitives. Compliance Rails are declarative, versioned documents that specify: (a) which regulatory requirements they address; (b) which Jasper protocol mechanisms satisfy each requirement; (c) which ClockChain event types constitute evidence of compliance; and (d) what audit reporting outputs the rail produces. Rails are published by GAPC or by authorized regulatory bodies in the GAPC Rail Registry.

Compliance Rails operate entirely above the core protocol stack. They do not modify Jasper wire formats, cryptographic requirements, or pillar specifications. They are additive — enabling compliance verification without altering protocol behavior. An agent **MAY** activate multiple Compliance Rails simultaneously, producing multi-jurisdictional compliance evidence from a single set of ClockChain records.

10.2 Available Compliance Rails (v1.0)

Rail ID	Jurisdiction / Framework	Key Mappings	Status
EU-AIAct-v1	EU AI Act (Regulation 2024/1689)	Risk tiers → Squirrel invariant severity levels; Art. 13 transparency → ClockChain URID + audit access; Art. 17 QMS → CGM Tier 2 + lifecycle events; Art. 9 risk management → SIS Operational Invariants	Ratified
US-NIST-RMF-v1	NIST AI RMF 1.0 (2023)	Govern → CGM Articles I, IV + responsible principal; Map → Capability Manifest + SIS; Measure → ClockChain audit logs + ORP; Manage → Conductor HALT/ESCALATE semantics	Ratified
ISO-42001-v1	ISO/IEC 42001:2023	Clause 6 (Planning) → Task Graph declaration; Clause 7 (Support) → GAPC Identity Registry; Clause 8 (Operation) → Squirrel + URIB; Clause 9 (Evaluation) → ClockChain + RFC 3161 anchoring; Clause 10 (Improvement) → SIS versioning + KEY_ROTATION events	Ratified
OECD-Principles-v1	OECD AI Principles (2024 update)	Principle 1.1 (Inclusive growth) → open URIB; Principle 1.2 (Human-centred) → Relational Invariants; Principle 1.3 (Transparency) → ClockChain + SIS	Ratified

Rail ID	Jurisdiction / Framework	Key Mappings	Status
		public declaration; Principle 1.4 (Robustness) → DOE determinism + HALT default; Principle 1.5 (Accountability) → CGM Article IV + responsible principal	
SG-AIFG-v1	Singapore Model AI Governance Framework 2.0 + Agentic AI Framework	11-principle AI Verify → Squirl invariant set templates; Agentic AI accountability → ClockChain + Conductor role; Human oversight → ESCALATE semantics + Observer Agent role	Ratified
G7-HIROSHIMA-v1	G7 Hiroshima Process Guiding Principles (2023)	Principle 1 (Trustworthy AI) → Squirl + ClockChain; Principle 3 (Labelling) → URID disclosure in user-facing outputs; Principle 7 (Technical standards) → Jasper v1.0 as reference implementation; Art. III Non-Maleficence + Operational Invariants	Ratified

10.3 Custom Rail Development

Organizations and regulatory bodies may develop, test, and publish custom Compliance Rails via the GAPC Rail Registry. The development process follows a three-stage pipeline:

28. **Draft Rail Submission:** The proposing organization submits a draft rail document to the GAPC Rail Registry, including: framework identification, normative requirements list, mapping table to Jasper primitives, and a test

suite of ClockChain scenarios that demonstrate compliance evidence generation.

29. **Community Review:** The draft rail undergoes a minimum 60-day public comment period. The GAPC Technical Steering Committee assigns a review working group. The RAP reviews for regulatory accuracy. Proposers respond to substantive comments with revision or rebuttal.

30. **Ratification:** Following review, the TSC votes on ratification. Ratified rails are published to the GAPC Rail Registry under a versioned identifier and become available for implementation.

10.4 Automated Compliance Reporting

Each active Compliance Rail defines a set of **ClockChain Query Templates** — structured queries against an agent's ClockChain that produce compliance evidence reports. The GAPC provides a reference implementation of a Compliance Report Generator that accepts a ClockChain export, a list of active rail IDs, and a reporting period, and produces a signed compliance evidence package in both machine-readable (JSON-LD) and human-readable (PDF) formats. This package may be submitted directly to regulatory authorities in jurisdictions that have recognized Jasper-based compliance evidence.

Automated compliance reporting reduces the burden on compliance teams from weeks of manual evidence collection to minutes of automated report generation — provided the agent's ClockChain records are complete, integer, and temporally anchored.

11. Global Adoption Roadmap

11.1 Phased Rollout

ROADMAP STATUS — As of 20 July 2026

Phase 1 is active. The GAPC Identity Registry, ClockChain public anchoring service, and reference implementations are available at

gapc.ai/jasper

. The Phase 1 pilot program is open to founding enterprise partners through Q4 2026.

Phase	Timeline	Name	Key Deliverables
Phase 1	Q3-Q4 2026	Foundation	Reference implementations (Python, TypeScript); GAPC Identity Registry launch; ClockChain anchoring service (clockchain.gapc.ai); pilot program with 10 founding enterprise partners; EU-AIAct-v1 and US-NIST-RMF-v1 rails ratified
Phase 2	Q1-Q2 2027	Ecosystem Build	URIB Relay Network operational; Jasper Gateway Adapters for AWS Bedrock, Azure AI, Google Vertex AI, Anthropic API, and OpenAI API; developer SDK in Go and Rust; Tier 1 and Tier 2 certification program launch; GAPC Developer Grants Program Year 1
Phase 3	Q3 2027-Q2 2028	Regulatory Integration	Formal MOU with EU AI Office; NIST NCCoE pilot program; Singapore IMDA partnership and SG-AIFG-v1 rail deployment; ISO/IEC

Phase	Timeline	Name	Key Deliverables
			JTC 1/SC 42 liaison submission; Tier 3 certification program launch with accredited third-party auditors
Phase 4	2028-2030	Global Standard	500+ certified organizations; 50+ jurisdictions with active compliance rails; UN-level recognition via UNESCO AI Ethics implementation pathway; Jasper v2.0 specification process initiated; GAPC membership exceeds 200 organizations

11.2 Adoption Incentives

- **Free Tier 1 Certification:** Open-source implementations of Jasper receive free Tier 1 certification from GAPC, including listing in the GAPC Certified Implementations Registry. No fees, no proprietary tooling required.
- **GAPC Developer Grants Program:** USD \$5 million annually from founding consortium organizations, distributed as grants of USD \$25,000–\$250,000 to developers, researchers, and startups building Jasper-compliant tools, adapters, and compliance rail implementations.
- **Regulatory Fast-Track Programs:** In partner jurisdictions (initially EU, Singapore, UK), organizations implementing Jasper at Tier 2 or above receive access to regulatory sandbox fast-track programs, reducing time-to-approval for AI deployments in regulated sectors.
- **Enterprise Consortium Benefits:** GAPC founding enterprise partners receive early access to draft specifications, working group participation rights, and co-branding opportunities on GAPC publications.

11.3 Success Metrics (KPIs)

Metric	Year 1 Target (July 2027)	Year 2 Target (July 2028)	Year 3 Target (July 2029)
Registered URID agents	10,000	250,000	2,000,000
Certified organizations	50	300	1,000+
Active compliance rails	6	20	50+
SDK downloads (cumulative)	100,000	1,000,000	5,000,000
Regulatory partnerships (MOU/formal)	3	10	25
Jurisdictions with active rails	8	25	50+
ClockChain blocks anchored/month	1,000,000	100,000,000	10,000,000,000
Enterprise deployments (Tier 2+)	25	200	500+

11.4 Call to Action

The Global AI Protocol Consortium invites organizations, regulatory bodies, developers, and researchers worldwide to join the Jasper community and help build the interoperability layer that global AI governance requires. There are four ways to engage:

31. Implement: Download the Jasper reference implementation and begin building compliant agents. Tier 1 certification is free for open-source implementations. Start at gapc.ai/jasper/start.

32. Participate: Join a Community Working Group. Working groups are open to all individuals and organizations. Current open working groups cover: identity, interoperability, safety invariants, compliance rails, and developer tooling. Apply at gapc.ai/working-groups.

33. **Partner:** Organizations deploying AI at enterprise scale are invited to become founding GAPC members, participate in the pilot program, and help shape Phase 2 of the roadmap. Contact .

34. **Regulate:** Regulatory bodies and government agencies are invited to engage with the Regulatory Advisory Panel, develop jurisdiction-specific compliance rails, and explore formal recognition of Jasper-based compliance evidence. Contact .

JOIN GAPC

All specification documents, reference implementations, and working group materials are available at gapc.ai

. This specification is published under the Open Protocol License v2.0 — free to implement, free to extend, free to critique.

12. Glossary

All terms defined in this section are capitalized throughout this specification to indicate their defined meaning. Terms used in lowercase carry their ordinary English meaning.

Agent ID

See *URID*. The persistent, cryptographically bound identifier for a Jasper-compliant AI agent.

BLAKE3

A modern cryptographic hash function used in Jasper for content-addressed operations, including ClockChain block chaining (prev_hash) and capability hash computation. Selected for high throughput and security margin.

CGM (Constitutional Governance Model)

Jasper's meta-governance layer — a six-article constitutional framework providing jurisdiction-neutral governance principles. Defined in Section 8.

ClockChain

A per-agent, append-only log of cryptographically signed, timestamped action blocks. Provides tamper-evident audit evidence of all agent actions. Defined in Section 5.

Compliance Rail

A versioned, declarative plugin document that maps the requirements of a specific regulatory framework to Jasper protocol primitives, enabling automated compliance evidence generation. Defined in Section 10.

Conductor Agent

An orchestrator agent role in a multi-agent system. Responsible for constructing and managing the Task Graph, delegating to Worker Agents, and handling failure conditions. Defined in Section 9.2.

Context Packet

A structured, signed payload carrying task context, authorization assertions, and ClockChain references between agents during a URIB interaction. Defined in Section 7.2.

Delegation Token

A scoped, time-limited, cryptographically signed authorization for one agent to act on behalf of another within a specified capability scope. Defined in Section 7.2.

Deterministic Zone

A portion of a Task Graph in which agent outputs must be reproducible given identical inputs. Contrast with Non-Deterministic Zone (NDZ).

DOE (Deterministic Orchestration Engine)

Jasper's multi-agent orchestration specification. Defines Task Graph semantics, agent roles, failure handling, deadlock prevention, and audit trail requirements. Section 9.

Ed25519

An elliptic curve digital signature algorithm (RFC 8032) used for all Jasper signing operations including ClockChain blocks, Jasper Envelopes, Squirrel Invariant Sets, and Capability Manifests.

GAPC (Global AI Protocol Consortium)

The issuing and governing body for the Jasper Universal Standards. Operates the GAPC Identity Registry, ClockChain anchoring service, Rail Registry, and governance working groups.

JCEL (Jasper Constraint Expression Language)

A structured, machine-readable expression language used in Squirrel Invariant Sets to formally specify invariant conditions and violation triggers.

Jasper

The Global AI Protocol — a universal interoperability and governance protocol for AI agents, defining five foundational pillars: ClockChain Identity Layer, Squirrel Personality Invariants, URIB Interoperability Fabric, Deterministic Orchestration Engine, and Constitutional Governance Model. The subject of this specification.

JFN (Jasper Federation Node)

An organization-operated URIB-compliant routing node that federates with the GAPC Relay Network, enabling organizational control over routing while maintaining global Jasper interoperability. Defined in Section 7.4.

JGA (Jasper Gateway Adapter)

An adapter that wraps a non-Jasper legacy API and presents it as a URIB-compliant endpoint, enabling legacy systems to participate in the Jasper ecosystem under a trust downgrade semantic. Defined in Section 7.5.

NDZ (Non-Deterministic Zone)

A declared zone in a Task Graph where agent outputs cannot be guaranteed to be reproducible (e.g., due to LLM sampling). NDZ actions require enhanced ClockChain logging including full payload hashes. Defined in Section 9.3.

Observer Agent

A read-only participant in a multi-agent orchestration that receives Context Packet copies and produces independent compliance audit logs. MUST NOT take actions or modify task state. Defined in Section 9.2.

ORP (Orchestration Replay Protocol)

The post-hoc audit procedure for reconstructing the complete execution of a Task Graph from combined ClockChain records for audit and incident investigation. Defined in Section 9.6.

SIS (Squirrel Invariant Set)

A versioned, cryptographically signed JSON-LD document declaring an AI agent's behavioral constraints across four invariant categories: Epistemic, Relational, Operational, and Stylistic. Defined in Section 6.3.

Squirrel

Jasper's behavioral constraint layer — a specification for declaring, signing, versioning, and auditing AI agent behavioral invariants. Not a safety system; a commitment transparency system. Defined in Section 6.

Task Graph

A Directed Acyclic Graph (DAG) of agent tasks with explicit dependency edges, timeout declarations, and failure semantics, declared by a Conductor Agent at orchestration initiation. Defined in Section 9.2.

Trust Handshake

A three-message protocol (CHALLENGE → ATTESTATION → ACKNOWLEDGMENT) through which two Jasper agents mutually verify each other's identity and invariant set before exchanging Context Packets. Defined in Section 7.2.

URIB (Universal Resource Interoperability Bus)

Jasper's cross-agent communication and capability exchange layer. Enables any two Jasper-compliant agents to interact, delegate, and audit regardless of vendor, model, or deployment environment. Defined in Section 7.

URID (Universal Resource Identity)

The per-agent identity primitive in Jasper, formatted as `jasper:///agent/{org-namespace}/{agent-name}/{version}#{capability-hash}`. Cryptographically bound to the agent's Ed25519 signing key pair. Defined in Section 5.2.

Worker Agent

An agent role that executes leaf tasks in a Task Graph under delegation from a Conductor Agent. Reports results and ClockChain references back to the Conductor. Defined in Section 9.2.

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